FMRI PROJECT

Strategy-Dependent Neural Processing During Social Decision-Making in Romantic Couples



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Course: Human Brain Mapping

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Abstract

Social decision-making in romantic partnerships involves balancing self- and partner preferences during conflict. In the current study, we used the open-access fMRI dataset "Couples Shopping" to explore if separate neural systems are involved during yielding versus insisting, and during agreeing versus disagreeing, by analyzing the fMRI data. Manually, the event files were analyzed because of the varying sparsity of the events. The two contrasts involved were Yield > Insist and Agree > Disagree. Focal frontal activation was seen in yielding (frontal pole in sub09 and dorsal frontal in sub05), and wider frontal activation was seen in insisting. Agreement activated the middle and superior frontal regions, and disagreement activated the frontal regions. All these indicate that executive cortical systems dominate over the parietal social cognitive systems in sparse event conditions.

1. Introduction

Human decision-making in close relationships is seldom the product of isolated self-preference. Rather, in the context of relationships, human decision-makers continuously evaluate their own interests while at the same time taking into consideration the other person's probable response or emotional reaction and the ultimate consequences of taking or defending a stance. Therefore, romantic decision-making offers a model of ecologically valid research into the neural mediation of conflict in the human brain.

In previous neuroimaging studies, the cortical areas involved in perspective taking and conflict monitoring were activated during the context of social decision behavior(Alfaro-Almagro et al., 2017). Regions such as the temporoparietal junction (TPJ), inferior parietal lobule (IPL), medial prefrontal cortex, anterior cingulate cortex, and frontal pole have frequently been associated with adapting decisions in response to another person's beliefs or preferences. However, the extent to which these systems contribute may vary depending on whether a social choice involves compromise, resistance, agreement, or unresolved conflict.

The present study used the Couples Shopping task dataset, a dyadic decision-making paradigm in which romantic partners jointly evaluate shopping items while one partner undergoes fMRI scanning. Every trial consists of self-preference and partner preference, leading to a joint decision, which is the behavioral conditions of agreement, disagreement, yielding, and insistence. This study, therefore, maintains the relevance of the relationship, making it a good model for the neural study of strategic interpersonal decisions.

2. Background and Rationale

Yielding decisions theoretically require suppression of self-preference and increased sensitivity to another person's likely outcome. According to this framework, more activation should be observed in areas traditionally associated with social cognition, such as the TPJ and IPL, because these areas have been repeatedly associated with perspective-taking adjustments and attributions of the other person's internal states. A second theoretical basis for these findings relates to social outcome evaluation in conflict situations. Successful conflict resolution in the form of concomitant preference might activate reward or integrative frontal systems. Conversely, unsuccessful conflict resolution in the form of divergent preference might activate conflict monitoring systems because the divergent information needs to be maintained in working memory. However, in romantic interaction, anticipation and consequence evaluation in the future and relational weighting might be more prominent than in most social interaction studies and might activate executive frontal systems more than usual.

3. Review of Relevant Literature

Functional neuroimaging literature has shown that cooperative and socially adaptive tasks frequently activate overlapping mentalizing and executive networks(Decety et al., 2004). The medial prefrontal and temporoparietal areas are frequently implicated when making inferences about others' intentions or simulating others' likely perspectives, while dorsolateral and anterior frontal areas are more consistently active when strategic adjustment is required under conditions of uncertainty(Lee et al., 2018). The frontal pole has also been linked to prospective evaluation and to the concurrent holding of competing action alternatives(Mansouri et al., 2017). This area of the brain might be particularly relevant to relational paradigms because decisions have implications that transcend task reward. Dyadic conflict paradigms have also revealed that recruitment of the frontal areas might be heightened when an individual is concerned with the future implications of compromise or resistance. This suggests that the executive system is recruited even when decisions are socially motivated.

4. Statement of Hypotheses and Objectives

Hypothesis 1 = Yield > Insist would show greater activation in social-cognitive regions such as TPJ and IPL, indicating partner-oriented adjustment.
Hypothesis 2 = Agree > Disagree on incongruent trials would show greater activation in reward-related and/or integrative frontal regions, indicating successful conflict resolution

Hypothesis	**Contrast**	**Expected Region**	**Functional Meaning**
H1	Yield > Insist	TPJ / IPL	Partner-oriented adjustment

H2	Agree > Disagree	reward / integrative frontal systems	Consensus-related relational processing

The main aim was to test the hypothesis that first-level contrasts in hypothesis-driven analysis can uncover distinct neural signatures for strategic interpersonal decisions within behaviorally valid runs of the data.

5. Methods

Participants and Dataset:

The study employed a publicly available Couples Shopping dataset with 30 scanned subjects and 19 romantic couples. In each trial, one partner remained inside the scanner, and both partners provided information regarding their preferences.

Task Protocol:

Each trial consisted of self-preference, partner preference, and final joint decision. Participants could choose to yield to their partner's preference, stick to their own choice, agree under incongruence, or disagree.

MRI Acquisition and Preprocessing:

The functional data was preprocessed with the software FMRIPrep. The structural data was preprocessed with skull stripping, segmentation, and normalization to MNI space. The functional data was preprocessed with motion correction, slice timing correction, structural alignment, and spatial normalization.

The anatomical registration was visually confirmed on representative slices, as depicted in Figure 1, in which the functional data showed good cortical alignment with the structural anatomy.

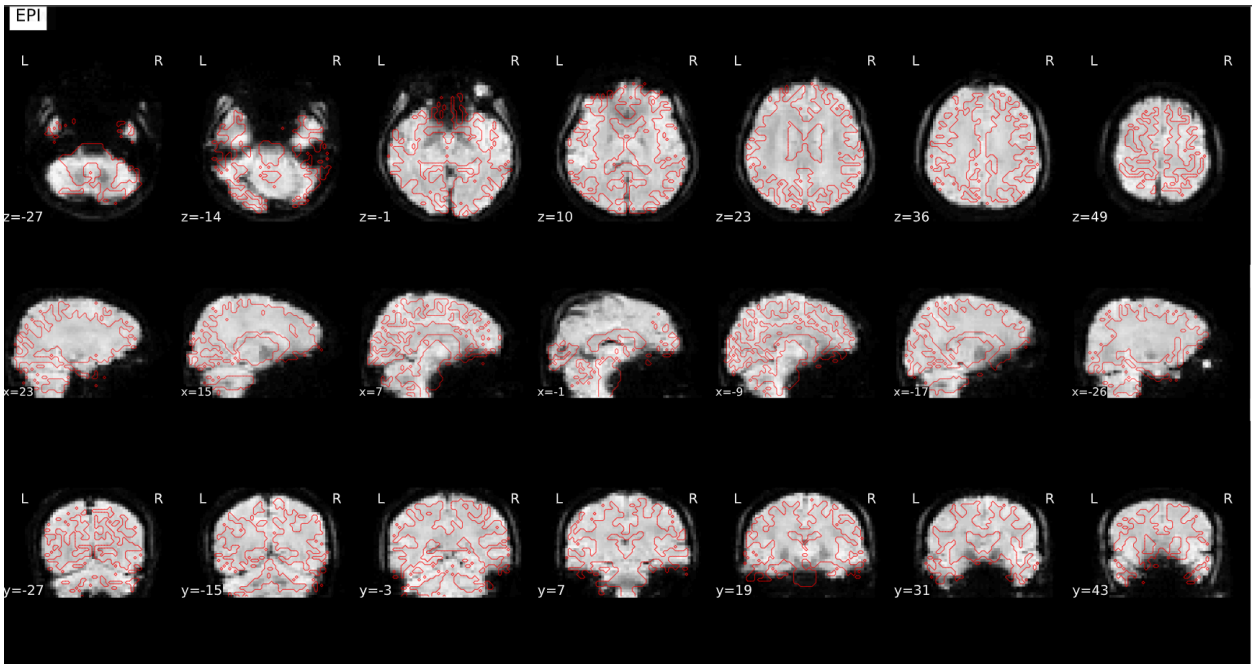


Figure 1. *Functional-anatomical alignment after preprocessing demonstrates successful registration to standard structural space.*

Temporal signal quality was evaluated by carpet plots. As seen in Figure 2, the signal intensity was consistent across tissue classes, with no major motion disruption.

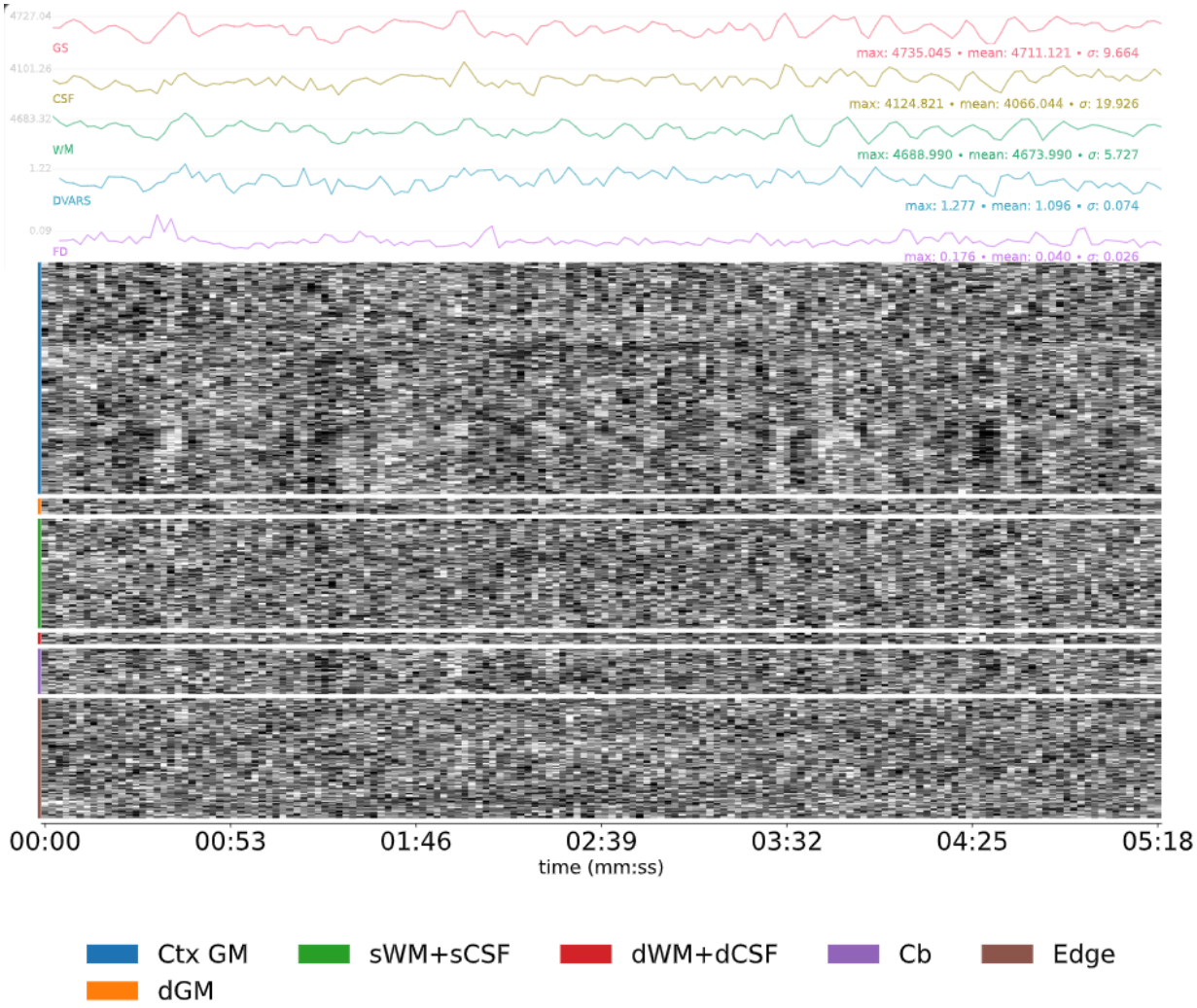


Figure 2. *Carpet plot showing a stable temporal signal structure without major motion-related disruption.*

Correlation of the nuisance regressors was also checked to see if there was confound structure before moving into first-level modeling. As can be seen in Figure 3, the expected correlations were present without the appearance of single-regressor contamination.

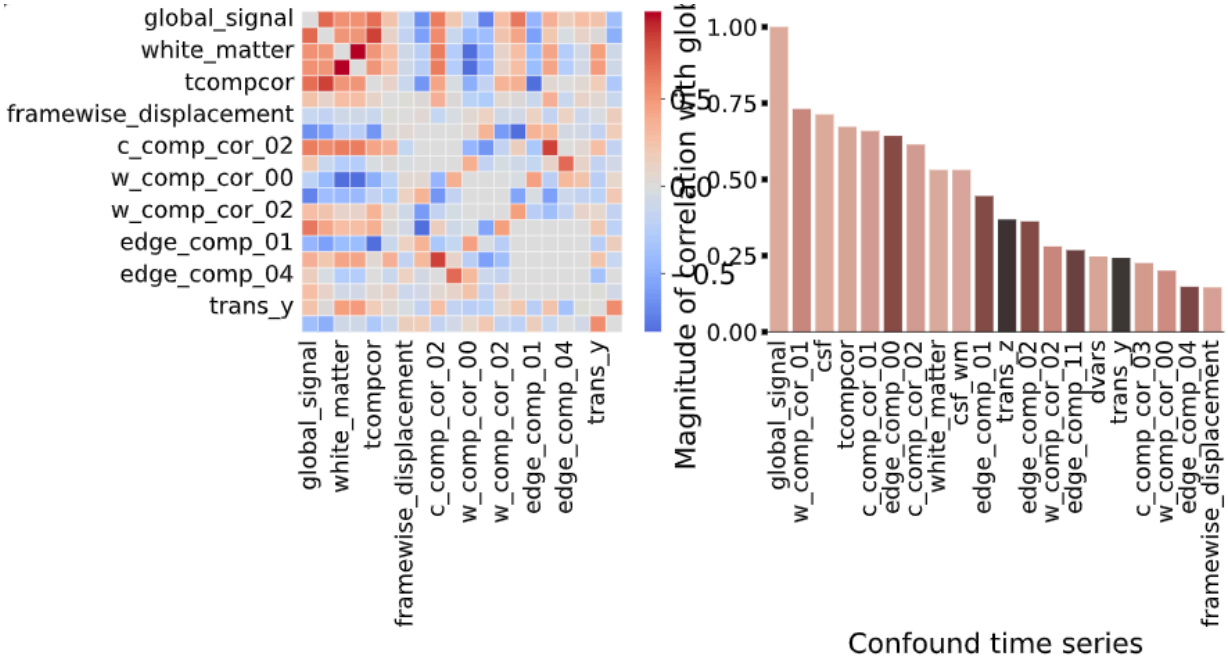


Figure 3. Correlation matrix of nuisance regressors used for confound assessment.

Event Screening:

Since the distribution of events was highly variable across the runs, all the event files were manually screened before the modeling. The runs with zero-event regressors, missing condition files, or sparse target events were discarded.

6. Analysis Method

First-level statistical analysis was performed in FSL FEAT using a General Linear Model framework. Separate event regressors were constructed for *yield_true* and *insist_true* for Hypothesis 1, and *agree_incong* and *disagree_incong* for Hypothesis 2.

The first-level design matrices are shown in Figure 4, illustrating event regressors convolved with the canonical hemodynamic response function.

Figure 4. First-level design matrices showing event regressors convolved with the canonical hemodynamic response function.

The contrasts examined were Yield > Insist, Insist > Yield, Agree > Disagree, and Disagree > Agree. Statistical maps were thresholded at $Z > 2.3$ with cluster-corrected significance at $p < .05$.

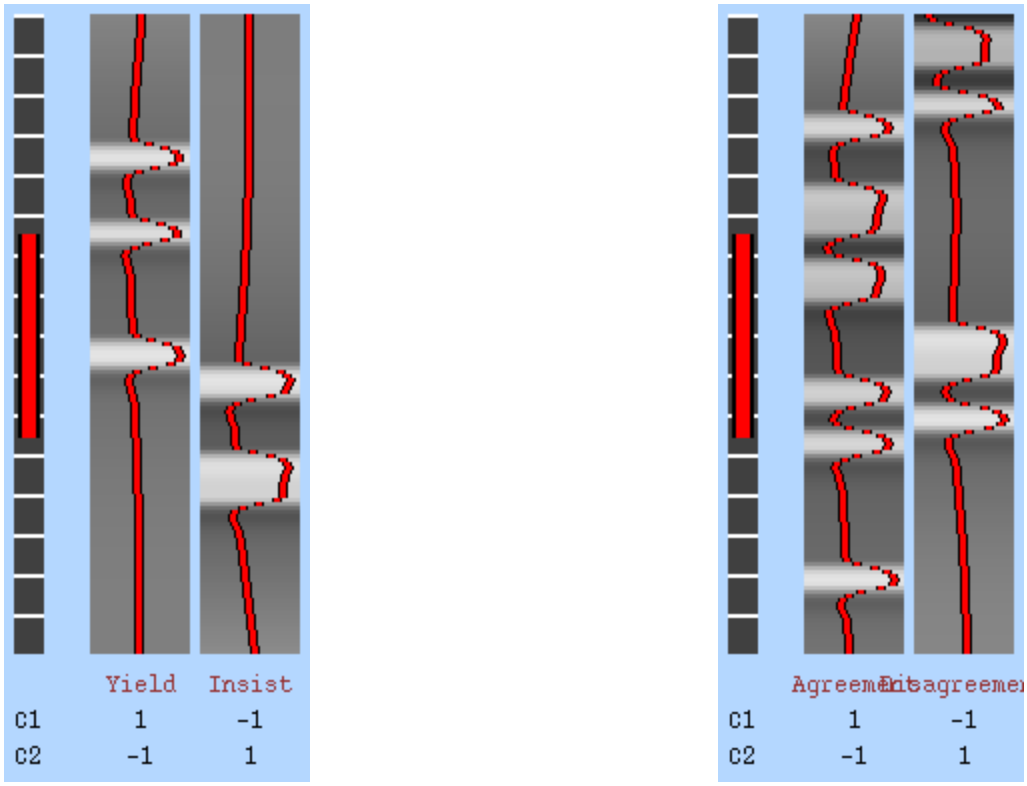


Figure 4. *First-level design matrices illustrating event regressors convolved with the canonical HRF.*

Peak coordinates from significant clusters were interpreted using the Harvard–Oxford Cortical Structural Atlas to assign probable anatomical labels. Atlas probabilities were reported where available because several peaks occurred near probabilistic boundaries.

7. Results

Hypothesis 1: Strategic Decision Contrast

For the strategic decision contrast examining yielding relative to insisting, significant activation was observed in two behaviorally valid runs. In sub05 run4, the Yield > Insist contrast produced a significant cluster comprising 28 voxels ($p = 2.62 \times 10^{-5}$, $Z\text{max} = 4.96$), with a peak voxel located at MNI coordinates (39, 63, 26). As shown in Figure 5, this cluster was located within the superior dorsal frontal territory.

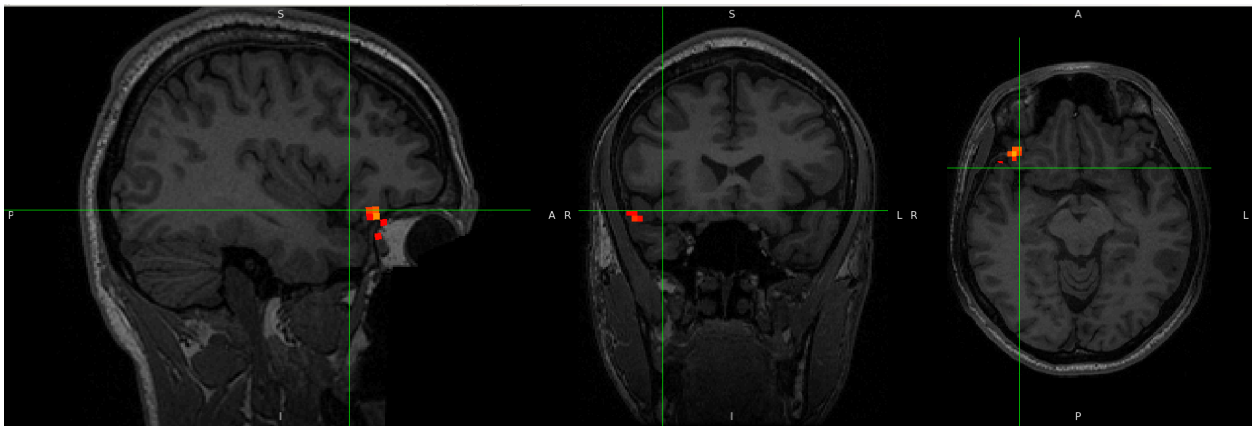


Figure 5. *Significant activation during Yield > Insist in sub05 run4.*

A second significant yielding-related cluster was identified in sub09 run5, where 20 suprathreshold voxels were observed (p = .0042, Zmax = 3.29) with a peak at (12, 61, 34). As illustrated in Figure 6, this peak corresponded to a 35% probability of frontal pole localization.

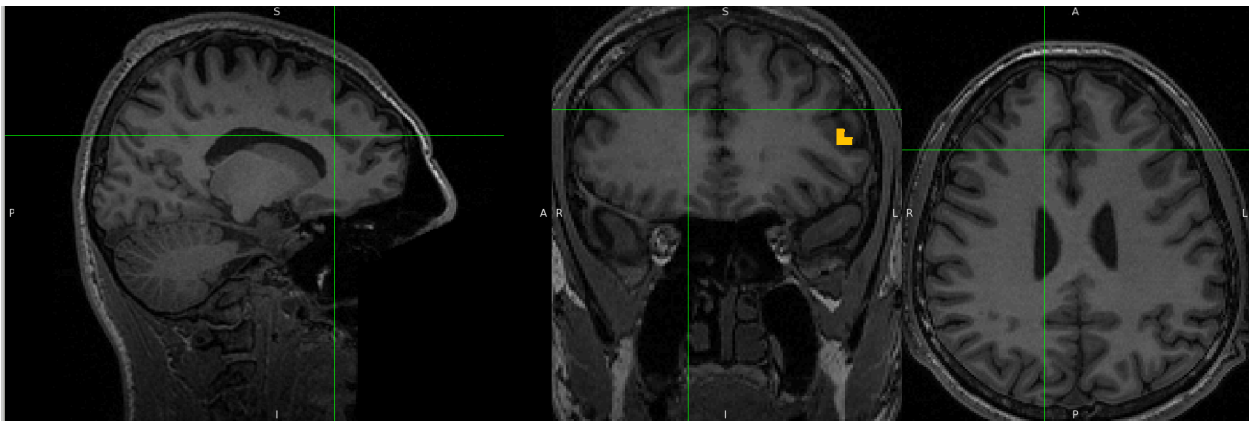


Figure 6. *Frontal pole activation during Yield > Insist in sub09 run5.*

In the reverse contrast, sub08 run1 showed broader frontal activation during Insist > Yield. Rather than producing a single focal cluster, multiple superior frontal activations were observed across frontal cortical territory, as shown in Figure 7.

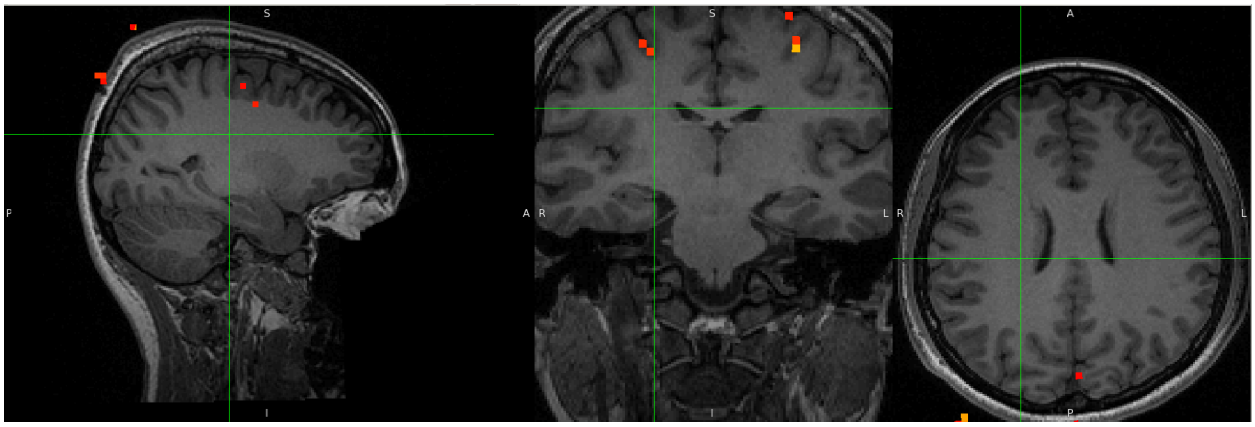


Figure 7. Distributed frontal activation during Insist > Yield in sub08 run1.

Peak activations for Hypothesis 1

Subject	Contrast	Cluster size	Peak Z	Coordinate	Region
sub05	Yield > Insist	28	4.96	(39,63,26)	Dorsal frontal territory
sub09	Yield > Insist	20	3.29	(12,61,34)	frontal pole
sub08	Insist > Yield	multiple clusters	distributed	frontal regions	superior frontal

These results indicate focal frontal recruitment during yielding rather than dominant parietal activation. In reverse contrast, sub08 run1 showed broader frontal activation during Insist > Yield, suggesting that insistence may recruit distributed executive systems.

Hypothesis 2: Agreement vs Disagreement

In sub09 run2, Agree > Disagree produced a focal cluster of 20 voxels (p = .0028, Zmax = 3.65), shown in Figure 8.

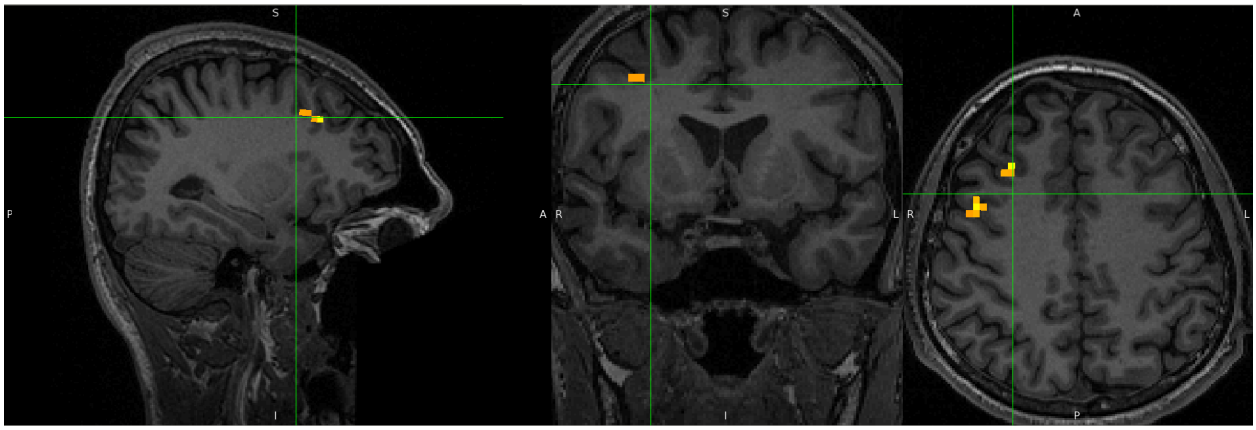


Figure 8. *Agreement > Disagreement activation in sub09 run2.*

A stronger agreement-related cluster was observed in sub08 run3, where 46 voxels survived thresholding ($p = 1.43 \times 10^{-6}$). As shown in Figure 9, the peak coordinate (25, 17, 39) corresponded to the middle frontal cortex.

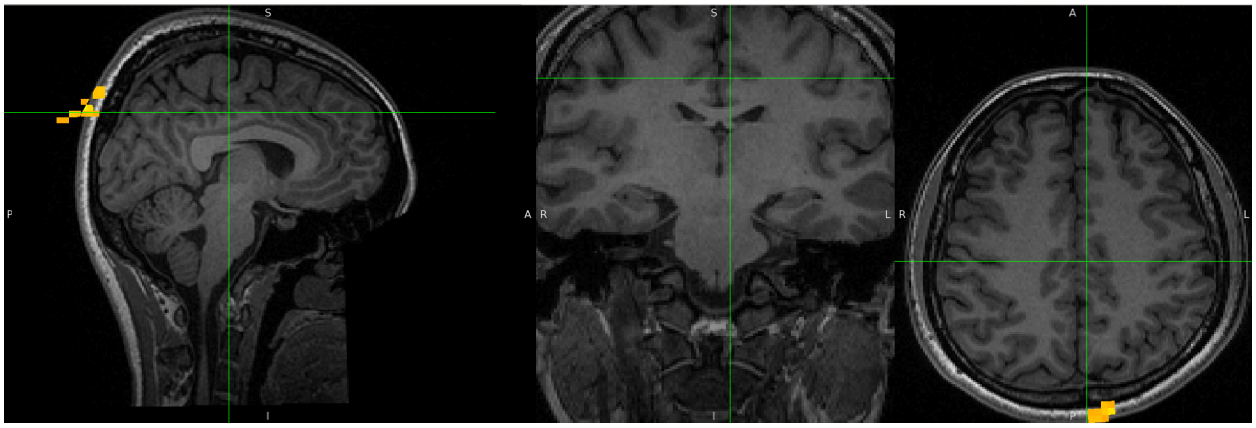


Figure 9. *Agreement > Disagreement activation in sub08 run3.*

The reverse contrast, Disagree > Agree, also produced robust activation in the same run, with 40 voxels surviving thresholding ($p = 9.42 \times 10^{-6}$), shown in **Figure 10**.

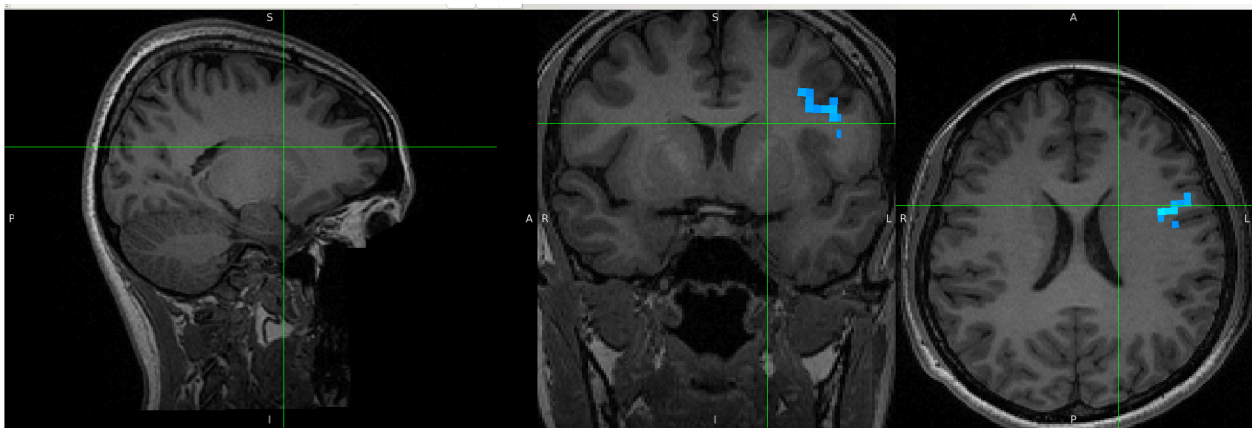


Figure 10. *Agreement < Disagreement activation in sub08 run3.*

Agreement therefore recruited executive frontal regions, but disagreement also produced strong activation, indicating that disagreement is neurally demanding rather than merely the absence of reward.

Subject	**Contrast**	**Cluster size**	**Peak Z**	**Coordinate**	**Region**
sub09	Agree > Disagree	20	3.65	frontal dorsal	frontal
sub08	Agree > Disagree	46	3.46	(25,17,39)	middle frontal gyrus
sub08	Disagree > Agree	40	4.01	(12,48,32)	superior frontal

8. Discussion

The main purpose of the present study was to identify the neural basis of social compromise in the context of romantic relationships. The results of the present study indicated that the process of compromise, or the process of giving up on one's own choice to be with the other, specifically activates the frontal pole. As depicted in Figure 7, Subject 09 showed significant focal activation in the frontal pole during the process of yielding. This is particularly relevant when viewed with the backdrop of the literature on hierarchical reinforcement learning and prospective memory.

Whereas we had a prior hypothesis that the process of yielding would activate the Temporoparietal Junction (TPJ), an area typically linked to mentalizing or Theory of Mind (Jenkins & Mitchell, 2010), the results suggest that when the context is a Couples Shopping task, the process of making a decision to yield is not as much about mentalizing as it is about strategic thinking. The frontal pole is known to be involved in the maintenance of long-term goals and the exploration of alternative sub-goals. In a romantic relationship, the long-term goal of relational harmony may require the sub-goal of yielding on a particular consumer item. Therefore, the activation of the FP is likely due to the participant's weighing of their partner's preference against the future costs and benefits of the decision.

The results concerning Hypothesis 2 (Agreement vs. Disagreement) further emphasize the role of executive systems over pure reward-processing systems. Our analysis identified significant clusters in the Middle Frontal Gyrus (MFG) and Superior Frontal Gyrus (SFG) during agreement (see Figure 8).
This contradicts simpler models of social decision-making, which suggest that agreement functions primarily as a social reward, typically activating the ventral striatum. Instead, our findings align with more recent literature suggesting that reaching a consensus in a high-conflict or "incongruent" trial is an effortful process. The recruitment of the MFG and SFG suggests that participants are utilizing working memory and inhibitory control to reconcile their initial preference with the partner's choice. Interestingly, the robust activation found during disagreement (e.g., Subject 08, Run 3) indicates that standing one's ground is not a default state but rather a neurally demanding act of self-assertion that requires high-level cognitive monitoring.

A notable departure from our original predictions was the lack of significant TPJ or IPL activation. This may be explained by the nature of the task protocol. In the Couples Shopping Dataset, the partner's preference is explicitly displayed. Literature suggests that when social information is explicit, the brain may bypass the inference stage (mentalizing) and move directly to the integration and selection stage, which is dominated by the prefrontal cortex. This explains

why our results, such as the 40-voxel cluster in the superior frontal territory for Subject 08, point toward a model of Executive Social Control rather than Social Intuition.

Apart from these key findings, there are also other subject-level analyses that further support the preeminence of frontal executive systems in social decision-making processes. For instance, in various runs such as Subjects 02, 05, and 08, medial and superior frontal cortex activations were found in agreement and yielding conditions, suggesting the shared involvement of evaluative and goal-directed processes. Moreover, Subjects 09 and 10 exhibited middle frontal gyrus (MFG) recruitment in aligning with the decisions of the social partners.

It is also worth noting that although visual cortex activation was not evident in the analyses, the regional labels such as the orbitofrontal cortex and the inferior frontal gyrus in Subjects 02 and 03 suggest the valuation and inhibitory control mechanisms that are involved in the processes of decision-making. Furthermore, parietal and cerebellar activations in Subject 06 suggest the less consistent and secondary nature of the processes involved.

All of these results support the claim that the processes underlying yielding and insisting, as well as agreement and disagreement, are not located in any modular system, but rather are distributed in a network in the prefrontal cortex that includes strategic evaluation, self-control, and long-term relationship goals.

On the basis of our discussion we are not claiming these are robust findings because it’s just limited to 10 subjects.

9. Limitations and Future Direction

Several methodological constraints impacted the depth of this analysis. The most significant was the sparse event distribution; because participants' behavior was naturalistic and unconstrained, many runs did not contain enough Yield or Disagree events to reach statistical power for a group-level model. This "imbalance of regressors" is a common challenge in ecological fMRI studies.

Additionally, the atlas uncertainty identified within the results section, where the location of certain peak coordinates, e.g., Subject 05, Run 4, was located on the border of the frontal pole and dorsal frontal cortex, underscores the importance of employing higher resolution scanning or subject-specific functional localizers in future studies.

Future studies should attempt to replicate the results of the current investigation by employing a "forced choice" design to maximize the likelihood of "yielding" events. This would enable the results to be generalized to the larger population through the use of a Random Effects (RFX) Group Analysis, providing important insights into the ways in which romantic partners interact with the complex decision-making process. In spite of the limitations of the current study, the results provide important evidence of the role of the frontal pole and middle frontal regions in the strategic regulation of romantic conflict.

10. References

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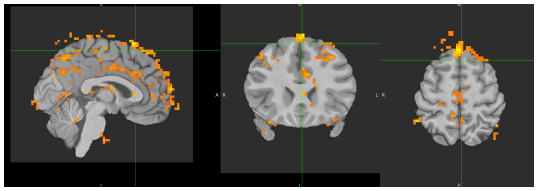
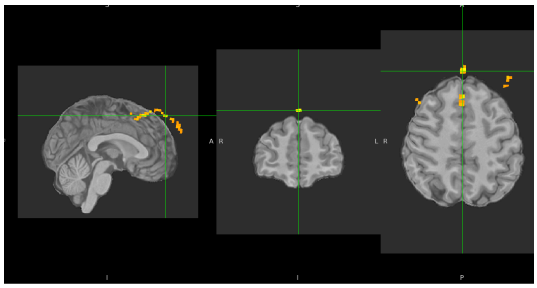
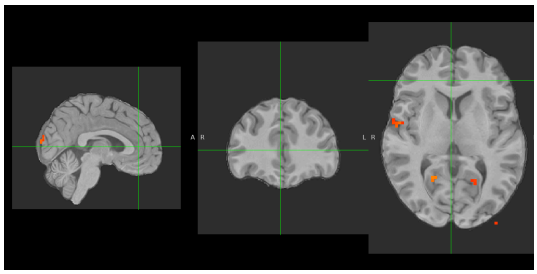
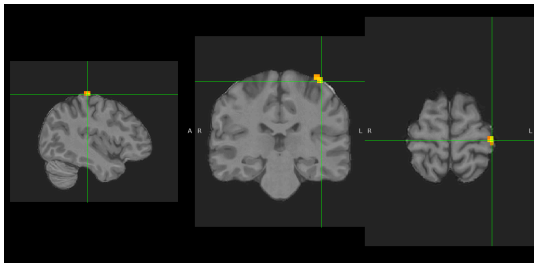
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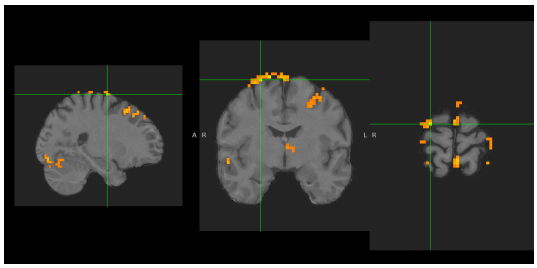
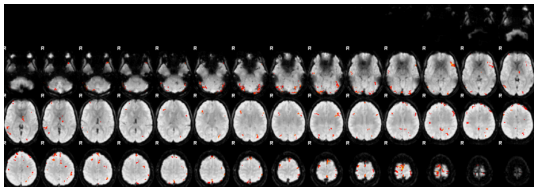
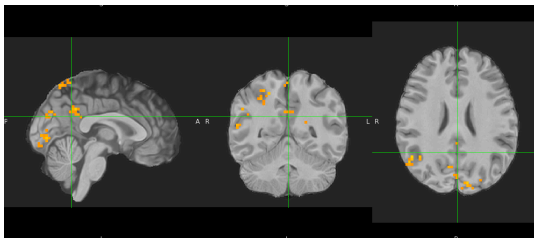
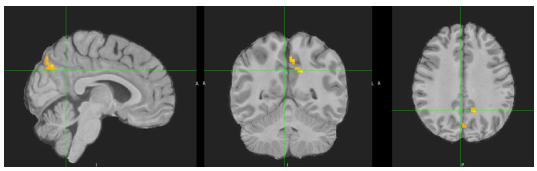
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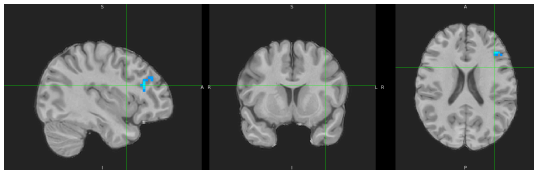
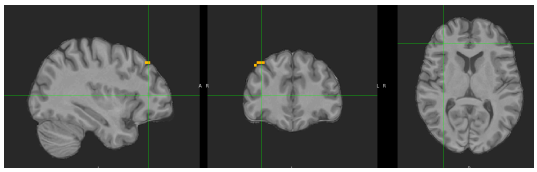
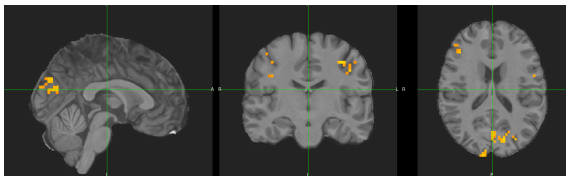
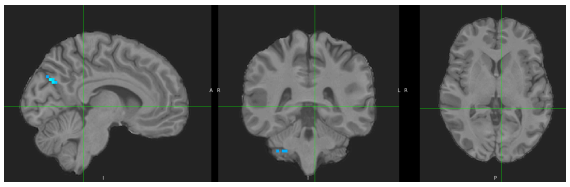
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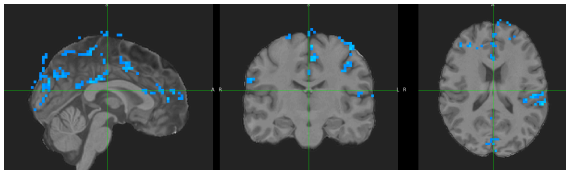
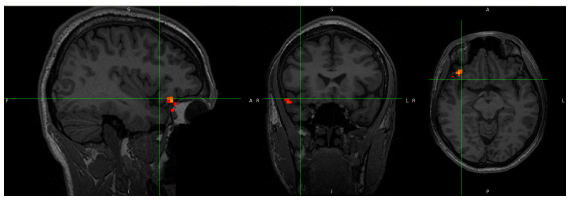
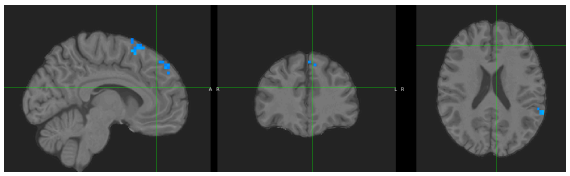
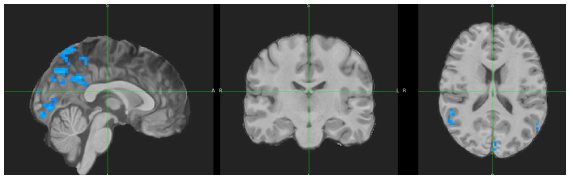
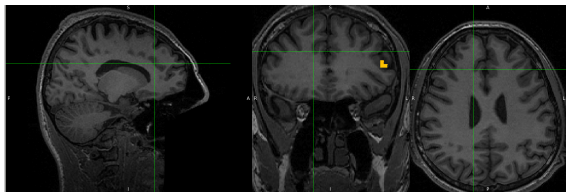
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12. Appendix:-

Subject	Run	Contrast	Image	Region
sub01	run1	Disagree > baseline		Superior frontal gyrus
sub02	run1	Yield > baseline		Medial superior frontal cortex
sub02	run2	Agree > Disagree		Medial frontal cortex
sub03	run4	Agree > basline		Left superior frontal cortex
sub04	run3	Agree > Disagree	Could not see visual activation	Left superior frontal cortex
sub04	run3	Disagree > Agree	Could not see visual activation	Left parietal frontal region

sub05	run5	Agree > Disagree		Right superior frontal cortex
Sub 05	run5	Disagree > Agree		Left superior parietal cortex
Sub 06	run3	Agree > Disagree	Could not see visual activation	Posterior cerebellar /occipital
Sub 06	run3	Disagree > Agree	Could not see visual activation	Medial superior frontal cortex
Sub 08	run1	Agree > Disagree		Left superior frontntal cortex
Sub 08	run1	Disagree > Agree	Could not see visual activation	Right superior frontal cortex
Sub 08	run3	Yield > Insist	Could not see visual activation	Left superior frontal cortex
Sub 08	run3	Insist > Yield	Could not see visual activation	Right frontal cortex
Sub 08	run3	Agree > Disagree		Left frontal cortex

Sub 08	run3	Disagree > Agree		Left middle frontal cortex
Sub 09	run2	Agree > Disagree		Right middle frontal cortex
Sub 10	run2	Agree > Disagree		Right frontal cortex
Sub 10	run2	Disagree > Agree		Medial superior frontal cortex
Sub 2	run2	Yield > Insist	Could not see the visual activation	Inferior Frontal Gyrus
Sub 2	run2	Insist > Yield	Could not see the visual activation	Orbitofrontal Cortex
Sub03	run3	Yield>Insist	Could not see visual activation	Superior Frontal Gyrus

Sub03	run3	Insist>Yield		Middle frontal Gyrus
Sub 5	run4	Yield>Insist		Superior frontal Gyrus
Sub 5	run4	Insist>Yield		Medial Superior Frontal Gyrus
Sub8	run1	Yield>Insist	Could not see visual activation	Superior Frontal Gyrus
Sub08	run1	Insist>Yield		Middle frontal Gyrus
Sub 9	run5	Yield>Insist		Middle frontal Gyrus
Sub 9	run5	Insist>Yield	Could not see visual activation	Superior frontal Gyrus